Abir Sarkar

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I am a Ph.D. student in Statistics and Data Science at **Cornell University**, with prior experience in quantitative model validation at **Capital One**. My research focuses on developing statistically principled methods in **active learning**, **stochastic optimization**, and **time series econometrics**.

Education

Cornell University 2025–2029
Ph.D. in Statistics and Data Science, Advisor: Prof. Martin T. Wells *Ithaca, NY*
Committee Member: Prof. Jelena Bradic

Indian Statistical Institute 2021–2023
Master of Statistics, specialization in Finance *Kolkata, India*

Indian Statistical Institute 2018–2021
Bachelor of Statistics *Kolkata, India*

Industry Experience & Internships

Capital One Jun 2023–Jul 2025
Senior Associate, Data Scientist, Model Validation *Bangalore, India*

- Led independent validation of enterprise-critical credit-risk models covering more than $140B in Commercial & Industrial, Commercial Real Estate, and Structured Finance exposure, including a $7B Municipal LGD model and approximately $20B in Healthcare C&I/CRE portfolios. Drove end-to-end technical challenge of model design, assumptions, performance, and implementation through independent replication, benchmarking, sensitivity analysis, and deployment validation.
- Worked across retail-fraud modeling, consumer-loss forecasting, deposit and cannibalization models, and CCAR/CECL stress testing, supporting high-stakes model-governance and regulatory reviews. Successfully completed two audit examinations.

JPMorgan Chase & Co. Summer 2022
Market Risk Analyst Intern *Mumbai, India*

Evaluated trading and risk models supporting foreign-exchange and commodities businesses within the Corporate & Investment Bank, with a focus on model performance, stability, and risk sensitivity. Developed a quantitative model-rating framework to strengthen ongoing performance monitoring and model governance.

Big Data Summer Institute, University of Michigan Summer 2021
Research Intern and Team Lead *Ann Arbor, MI*

Led a research team on Michigan Medicine clinical data, developing and benchmarking missing-data imputation pipelines spanning classical statistical methods and machine learning. Built and evaluated predictive models for hospitalization outcomes, translating incomplete clinical data into reliable downstream prediction.

Submitted Articles & Ongoing Projects

Minimax Optimality of Shampoo for Nonstationary Stochastic Optimization
Preprint, Submitted to AISTATS
Abir Sarkar[*], Nirmalya Mandal[*], and Martin T. Wells
[*]Equal contribution.
Established high-probability tracking bounds for Shampoo under evolving objectives, revealing explicit momentum–noise and stability–geometry tradeoffs and proving asymptotic minimax optimality over causal optimization procedures.

Adapt or Forget: Provable Tradeoffs between Adam and SGD in Nonstationary Optimization
Preprint, Submitted to ICLR
Sharan Sahu[*], **Abir Sarkar**[*], Cameron J. Hogan[*], and Martin T. Wells
Extended Adam's analysis to changing stochastic objectives, explicitly accounting for both first-moment estimation error and perturbations from its adaptive preconditioner. Derived hyperparameter-dependent bounds showing how these two components affect the tradeoff between reducing gradient noise and tracking objective drift, with corresponding comparisons to SGD.

Sequentially Balanced Active Inference
Ongoing Project
Abir Sarkar and Jelena Bradic
Developed a protected plug-in Neyman allocation that continues sampling regions even when initial labels underestimate their prediction errors. Accounted for the changing regional moments induced by sampling without replacement and bounded cumulative excess variance relative to the optimal allocation for the remaining population by a logarithmic term.

Double Local-to-Unity: Inference under Nearly Nonstationary Volatility

Preprint, Submitted to Journal of Econometrics

Abir Sarkar and Martin T. Wells

Extended moderate-deviation autoregressive theory to jointly near-unit-root mean and stochastic-volatility dynamics, allowing volatility moments to diverge. Identified the required volatility normalization and obtained Gaussian and self-normalized Cauchy limits in the mildly stationary and mildly explosive regimes, respectively.

Empirical Bayes Predictive Density Estimation under Covariate Shift in Large Imbalanced Linear Mixed Models

Preprint, Submitted to JASA

Abir Sarkar, Gourab Mukherjee, and Keisuke Yano

Developed a data-fission method for estimating predictive Kullback–Leibler risk when replication is scarce and future covariates differ from observed covariates. Accounted explicitly for dependence caused by sample reuse, deriving replication-dependent risk bounds and guarantees relative to the best predictive density in the specified prior class.

Publications

Is There an AI Bubble? Robust Date-Stamping for Periods of Exuberance

Frontiers of Mathematical Finance 10, 1–42 (2026)

Abir Sarkar and Martin T. Wells

Extended recursive bubble detection to asset prices with highly persistent stochastic volatility, where homoskedastic tests can mistake volatility fluctuations for explosive behavior. Derived distinct, nuisance-parameter-free calibration rules for bubble origination and collapse, improving detection and dating accuracy in simulations.

Asymptotic Breakdown Point Analysis for a General Class of Minimum Divergence Estimators

Bernoulli 32(1), 698–722 (2026)

Subhrajyoty Roy, **Abir Sarkar**, Abhik Ghosh, and Ayanendranath Basu

Extended breakdown-point theory for minimum S-divergence estimators to general parametric models, deriving dimension-independent contamination-tolerance bounds that encompass density power divergence estimators.

Teaching Experience at Cornell University

ECON 2801: Game Theory—For Economics, Politics, Knowledge, and Rationality — Fall 2025, 2026
Grader — *Instructor: Prof. Kaushik Basu*

STSCI 4981: Sports Analytics in Practice — Spring 2026
Teaching Assistant and Grader — *Instructor: Prof. Martin T. Wells*

STSCI 5080: Probability Models and Inference — Fall 2026
Teaching Assistant and Grader — *Instructor: Prof. Kengo Kato*

Conference Presentations

- **Is There an AI Bubble? Robust Date-Stamping for Periods of Exuberance**
  - The 39th New England Statistics Symposium (2026), *From Data to Discovery: Statistical Insight in the Age of AI*.
  - Invited Talk, Department Colloquium, University of Connecticut, Storrs, CT (2026).
- **Empirical Bayes Predictive Density Estimation under Covariate Shift in Large Imbalanced Linear Mixed Models**
  - Annual Conference of the International Indian Statistical Association, Kochi, India (2024).
  - International Conference on Statistics and Data Science, Seville, Spain (2025).

Awards and Achievements

- **Director's Award**, Capital One, for contributions to the validation of Commercial C&I and Municipal default models.
- **Statistically Significant Award**, Capital One, for exceptional performance across multiple projects under demanding timelines.
- **INSPIRE Scholarship**, Department of Science and Technology, Government of India, 2018–2023.
- Secured **1st rank** in the **Jagadis Bose National Science Talent Search (JBNSTS)**, 2018.
- Ranked **3rd** in both the B.Stat. (2018–2021) and M.Stat. (2021–2023) cohorts at the Indian Statistical Institute.
- Ranked **4th** and **5th** in the **Simon Marais Mathematics Competition** in 2018 and 2019, respectively.

Technical Skills

Programming: Python, R, SQL, C++, LaTeX, Git

Languages: English, Hindi, Bengali